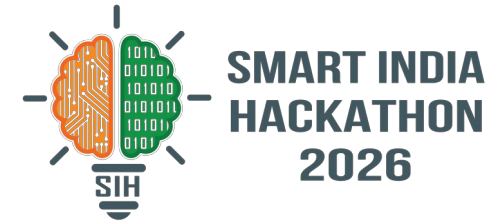
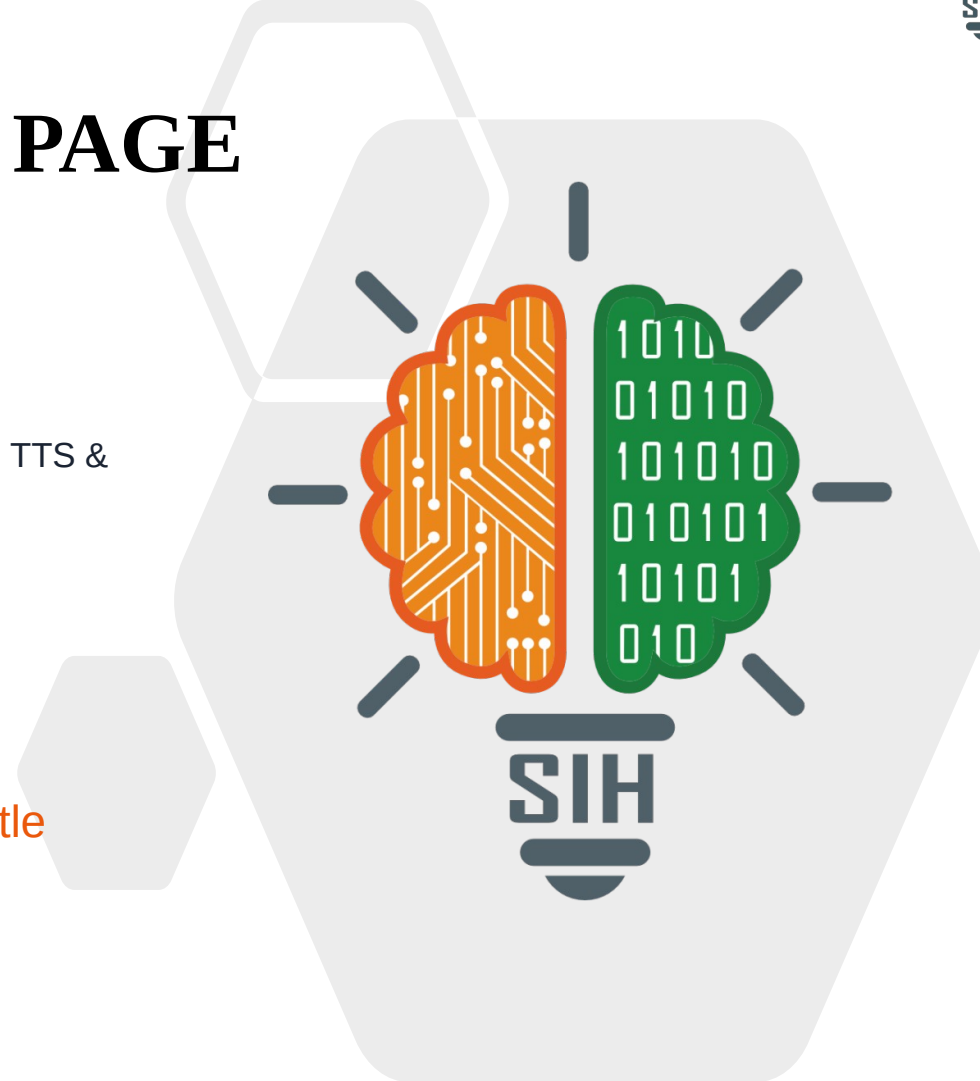


SMART INDIA HACKATHON 2026



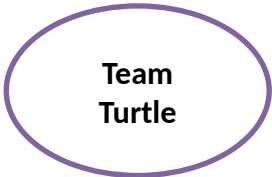
TITLE PAGE

- **Problem Statement ID** – SIH26173
- **Problem Statement Title**- iTantra – Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 188881
- **Team Name (Registered on portal)** Team Turtle

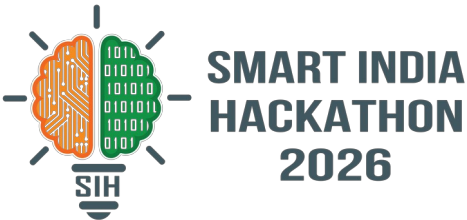


iTantra

Speak. Send text. Hear it back — offline.



iTantra: Speak. Send Text. Hear It Back — Offline

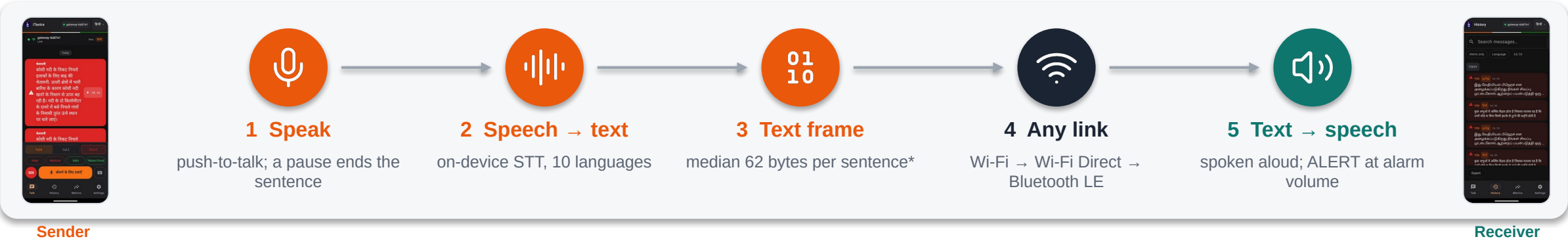


Proposed Solution (Describe your Idea/Solution/Prototype)

An Android app that turns speech into text on the sender's phone, sends only the text, and speaks it aloud on the receiver's phone. **No tower, no internet, no cloud.**

Demo film
1 min 55 s
itantra-106.pages.dev/demo





Detailed explanation of the proposed solution

-  10 Indian languages, speech and voice, fully on the phone
-  ALERT plays at alarm volume; one-tap SOS sends GPS/NavIC location
-  Push-to-talk walkie-talkie and hands-free call mode

How it addresses the problem

Raw audio	256k
Opus voice	12k
iTantra text	165

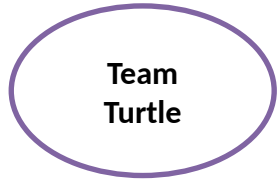
bits per second, one spoken sentence (log scale)

Output is still speech — people who cannot read are included

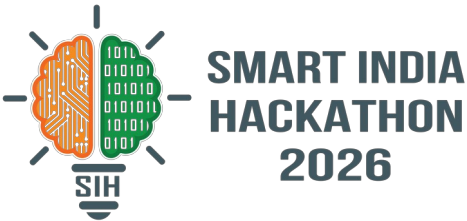
Innovation and uniqueness of the solution

-  Speech → text → speech used as the codec for thin links
-  Alerts re-voiced on arrival, cannot be interrupted
-  Auto link fallback, one packet format on every link
-  LITE profile switches on by itself below 6 GB RAM

*Median over 310 FLEURS sentences in a unit test; excludes encryption, alert signature and link headers. Bars assume a ~3 s sentence.



TECHNICAL APPROACH



Technologies to be used (e.g. programming languages, frameworks, hardware)

Android app
Kotlin · Jetpack Compose

Voice detection
Silero VAD

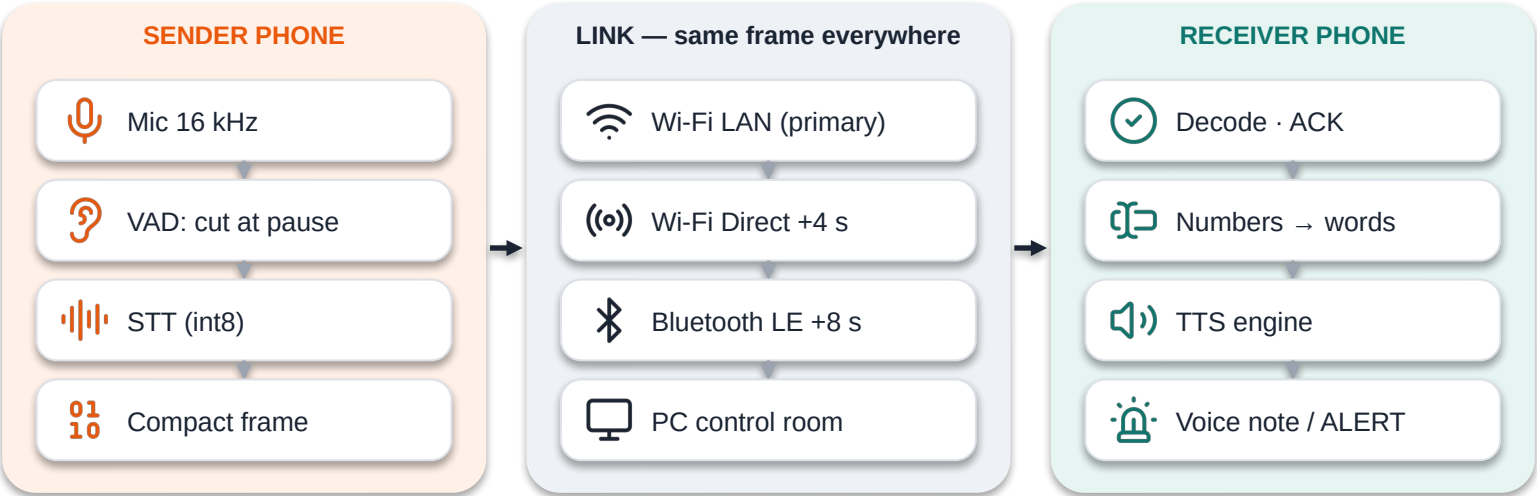
Speech-to-text
SraVaani-1.0 · int8 · sherpa-onnx

Text-to-speech
VITS Rasa · Indic-Mio

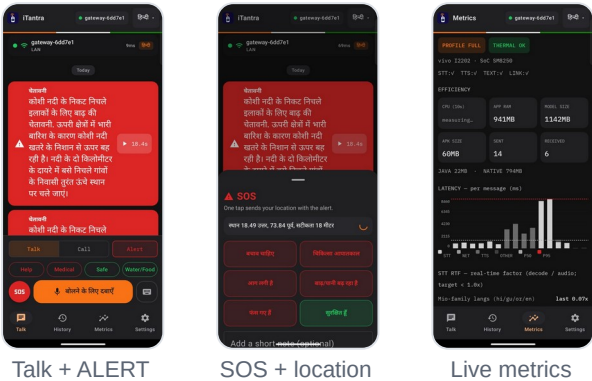
Links
Wi-Fi (mDNS/TCP) · Wi-Fi Direct · BLE

Hardware
Any Android phone · PC control room

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype — real screens



BUILT & RUNNING

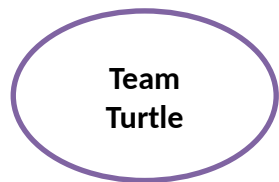
Push-to-talk, call mode, 10-language STT/TTS, ALERT, SOS + location, LITE profile, compact frame, PC control room

BUILT · TEST PENDING

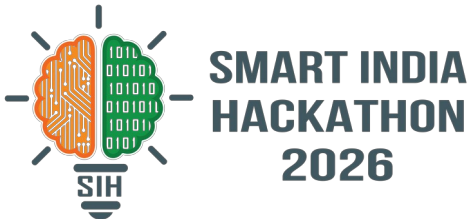
Wi-Fi Direct and Bluetooth LE fallback, phone-to-phone

PLANNED

Smaller voices (hi/gu/or/en), per-language model download, encrypted SOS location, ESP32 radio bridge



FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

Measured in the running prototype (in-app logging)

20×
faster than speech
STT real-time factor 0.05

0.4–0.9 s
speech end → text ready
Snapdragon 870

688 MB
app RAM, LITE profile
≈17% of a 4 GB phone

0.3%
idle CPU, LITE profile
1.6% in FULL

5.9%
word error rate
preliminary: 10 clips

Test phone: Snapdragon 870, 12 GB RAM, LITE profile forced. On 4–6 GB phones LITE switches on by itself; its 688 MB footprint does not depend on the phone's RAM. Speed on budget chips and end-to-end latency: measurement in progress.

Potential challenges and risks

Strategies for overcoming these challenges

Phones with 3–6 GB RAM	→ LITE profile auto-selects below 6 GB: 2 threads, no live captions, engines unload after 60 s
1.14 GB of model files	→ Per-language on-demand download and smaller voices for hi/gu/or/en (planned)
Wi-Fi Direct / Bluetooth not yet tested phone-to-phone	→ Two-phone tests on real devices; normal Wi-Fi stays the primary link
Noise, accents, code-mixing raise errors	→ Keyword boosting for alert words; 30-clip-per-language evaluation; typed text as fallback

Viability

No airtime, server or cloud fees

Open models: MIT · Apache-2.0 · CC-BY-4.0

One packet format on every link

Potential impact on the target audience



Disaster response teams

Voice coordination when towers and internet are down



Coastal fishing communities

Spoken local-language alerts over weak links



Low-literacy users

Hear an alert a text message would miss



Remote field staff

Offline voice link in poor-coverage areas

≈1,500×

less data than raw audio*

1.03 s

ALERT arrival → spoken aloud
(measured, Wi-Fi)

10

Indian languages, speech in and out, on the phone

Benefits of the solution (social, economic, environmental, etc.)



Social

Alerts spoken aloud in Indian languages — usable by people who cannot read



Economic

No airtime, cloud or licence fees; runs on phones people already own



Environmental

Fewer bytes per message means less radio airtime (battery gain not yet measured)

*62 B per ~3 s sentence vs 256 kbps raw audio; 62 B is a unit-test median. No field pilot yet. Photos (Wikimedia Commons): NDRF – Ministry of Home Affairs, GODL-India; fishermen – Saral Shots, CC0; woman on phone – Ammalu, CC BY-SA 3.0; rail workers – David Brossard, CC BY-SA 2.0.

Details / Links of the reference and research work



Open models (all run on the phone)

SraVaani-1.0 speech-to-text · ARTPARK, IISc · MIT
huggingface.co/ARTPARK-IISc/SraVaani-1.0

Indic-Mio text-to-speech · SPRING Lab, IIT Madras · Apache-2.0
huggingface.co/SPRINGLab/Indic-Mio

VITS Rasa 13 text-to-speech · AI4Bharat · CC-BY-4.0
huggingface.co/ai4bharat/vits_rasa_13

Silero VAD voice detection · MIT
github.com/snakers4/silero-vad



Runtime and evaluation data

sherpa-onnx on-device ASR/TTS runtime · Apache-2.0
github.com/k2-fsa/sherpa-onnx

Google FLEURS speech clips (STT evaluation)
huggingface.co/datasets/google/fleurs



Android platform guides

Wi-Fi Direct (Wi-Fi P2P)
developer.android.com/develop/connectivity/wifi/wifi-direct

Bluetooth and BLE GATT
developer.android.com/develop/connectivity/bluetooth

Audio focus (ALERT playback)
developer.android.com/media/optimize/audio-focus



Our work

Prototype code and measurements
github.com/prasadxa/itantra

Project website: overview, results, APK
itantra-106.pages.dev

Demo film, 1 min 55 s, captioned
itantra-106.pages.dev/demo



Website

[overview](#) · [results](#) · [APK](#)



Demo film

1 min 55 s

Licences as stated on each model card. No cloud API is used for speech recognition or speech synthesis.